Patch-Based Diffusion Models for Solving Inverse Problems in Computed Tomography Imaging

Executive Summary

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Purpose and Motivation

Computed tomography (CT) reconstructs anatomy from X-ray measurements [1]. Fewer angles reduce acquisition time and radiation exposure but make reconstruction underdetermined, causing streaks or blur without an image prior [2].

Diffusion models provide expressive learned priors [3, 4], but whole-image models can be data- and memory-intensive. Hu et al. proposed the Patch Diffusion Inverse Solver (PaDIS) [5], which combines position-labelled patch scores from changing partitions to avoid seams. They reported that PaDIS consistently outperformed a comparable whole-image diffusion model on sparse-view CT.

This work tested whether that result could be reproduced on a public dataset with a specified fan-beam geometry, and which choices drive performance. Independent testing matters because hidden implementation choices hinder validation, making this a reproduction and extension rather than a numerical rerun.

Approach

Training, tuning and evaluation were reimplemented within LION [6, 7] rather than treating the released code as executable ground truth. Patient-separated LIDC-IDRI slices provided the data [8, 9]. The default subset contained at most four slices per patient, while an ablation used every processed slice. A fully specified two-dimensional fan-beam operator, implemented through LION's tomosipo and ASTRA stack [10, 11], simulated noise-free measurements, isolating angular undersampling from photon-count effects.

Because the original PaDIS paper and released code differ, three tracks capture alternative interpretations. Paper-Faithful follows the published equations and sampling choices [5], while Repo-Faithful follows the repository and reproduces an adapted fork's outputs within sampler variation on matched inputs [12]. LION-Physical uses the same priors but normalises data-consistency gradients using the squared norm of the composed CT operator, avoiding geometry-specific calibration constants [13, 14].

This work compares classical, whole-image diffusion and PaDIS reconstruction, including FDK [15], Chambolle–Pock TV [16, 17] and PnP-ADMM [18]. Main tests use 8 and 20 full-angle views, with extensions to 60 views, limited angles and native 512×512 resolution. Ablations vary patch size, training-set size, positional encoding, initialisation and noise schedule. Reported measures include PSNR, SSIM [19], MAE, measurement residuals and image-level bootstrap intervals [20].

Principal Findings

The reconstructions in Figure 1 show representative fan-beam results from PaDIS-VE-DPS and the comparison methods. On the exploratory four-image native 512×512 subset (Figure 2), LION-Physical PaDIS-VE-DPS achieved 40.08 dB PSNR and 0.960 SSIM, exceeding FDK and CP-TV without requiring a whole-image model at that resolution. The separately tuned PaDIS-VE-DDNM configuration [21] produced the best 20-view and 60-view PSNR values in this work, reaching 36.49 and 41.55 dB respectively, while the LION-Physical VE-DPS rows [22] reduced relative sinogram residuals to approximately 10^{-3} and

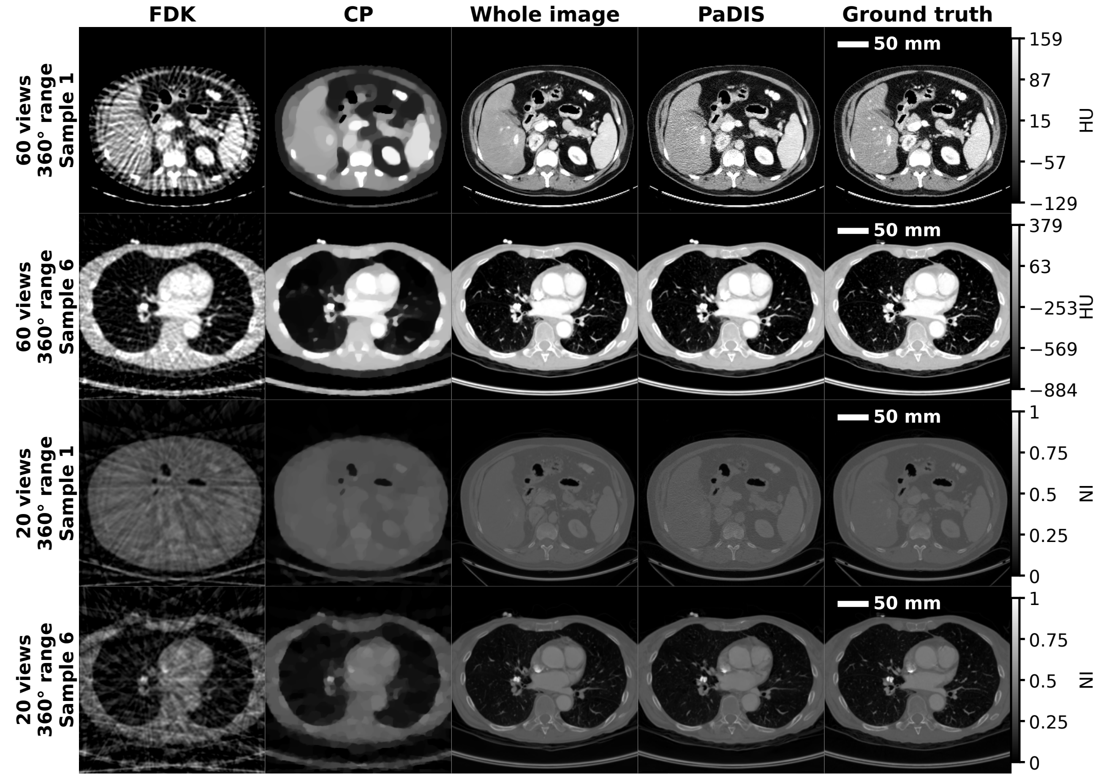

Figure 1: 60-view HU (top) and 20-view NI (bottom) reconstructions comparing FDK, CP, whole-image VE-DPS (labelled Whole image) and PaDIS-VE-DPS (labelled PaDIS).

therefore kept the outputs closely tied to the measurements.

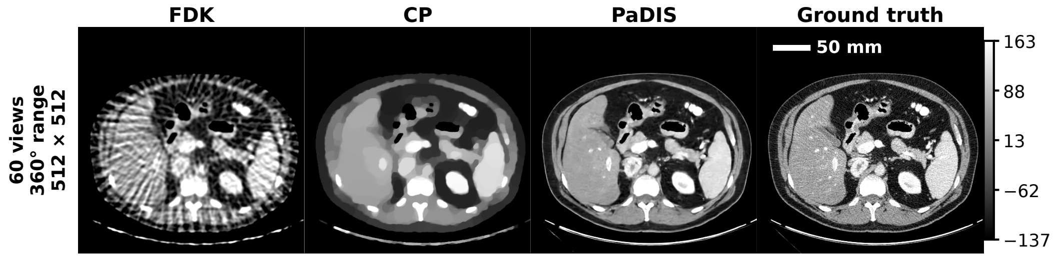

Figure 2: Native-resolution 60-view PaDIS-VE-DPS reconstruction, labelled PaDIS and displayed in HU.

We did not reproduce the stronger claim of consistent PaDIS superiority. Under VE-DPS, the whole-image model achieved 34.25 dB at 20 views and 29.58 dB at 8 views, compared with 32.82 and 27.65 dB for PaDIS-VE-DPS, and we observed the same reversal in the Paper-Faithful track. PaDIS exceeded the whole-image result only for some separately selected sampler configurations, notably VE-DDNM, so these comparisons do not isolate the prior alone. Patch training is therefore viable and sometimes advantageous, but the result depends on the interaction between prior, sampler and forward-operator scaling.

The ablations changed the interpretation further. A 96×96 patch model had almost the same mean PSNR as the whole-image model, while intermediate smaller patches performed worse, suggesting that global context remained valuable. However, the non-default patch and no-position models inherited the default PaDIS-VE-DPS setting and were not individually optimised. Positional channels improved measurement-conditioned PSNR by only 0.3 to 0.4 dB, in contrast to the failure reported in the original PaDIS paper, and FDK and Gaussian initialisation converged to similar final quality. The geometric noise schedule instead improved PSNR by about 1.3 dB over the released code's EDM-style default. Using all available training slices did not improve PaDIS-VE-DPS under a fixed training-time budget, although fewer effective epochs mean this is not evidence that additional data are intrinsically unhelpful.

Among the four untuned samples per method in Figure 3, random PaDIS partitions avoided the rectangular boundaries created by fixed patch stitching, although this small set cannot rank distributional quality. We recorded roughly two minutes per PaDIS-VE-DPS reconstruction, compared with about 2.6 minutes for whole-image VE-DPS and 24 minutes for fixed averaging or stitching, but concurrent jobs prevent an isolated speed comparison.

Reproducibility, Limitations and Implications

Exact reproduction of the original PaDIS paper was not possible. Important architecture information was distributed inconsistently between the paper, its cited Patch Diffusion construction and code [5, 12, 23]. Training patch counts, validation and checkpoint selection, test slice identities, complete physical geometry, and the hyperparameter search protocol were not specified. The original PaDIS paper and released code form a combined but internally inconsistent specification, containing materially different algorithmic choices [5, 12]. By using three tracks, we made these differences explicit and showed that they can change scientific conclusions. The reversal of the central comparison is therefore an informative reproducibility result rather than a failed outcome.

This work remains limited to one dataset, simulated two-dimensional measurements and a small test set. Noise-free sinograms omit photon noise, scatter, beam hardening and motion, so we do not establish clinical low-dose performance. Some expensive experiments used only four images, confidence intervals do not include retraining variability, and no radiologist assessed lesion preservation or hallucination. A larger native-resolution test set is therefore needed before drawing a population-level comparison between reconstruction methods.

The practical conclusion is that PaDIS should be viewed as a memory-scalable route to high-resolution priors, not as a universally superior replacement for whole-image diffusion. Future work should use equal-compute and equal-epoch comparisons, multiple training seeds, calibrated measurement noise, external datasets and clinically relevant task assessment for safe clinical deployment. This work also demonstrates that inverse-problem papers should publish complete geometry, data splits, tuning procedures and code that matches the stated algorithms. Otherwise, apparent method improvements may instead reflect hidden implementation choices. The complete project codebase is available here, and the training runs and saved models are available here.

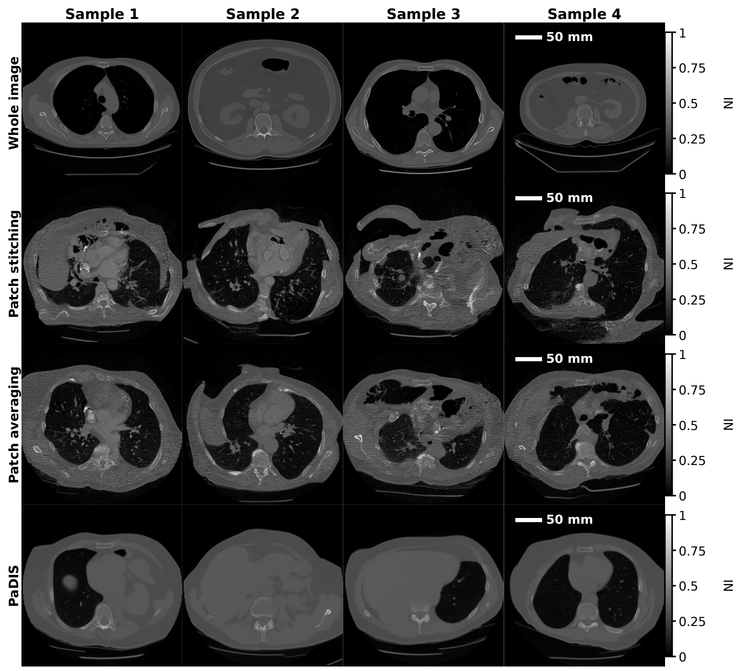

Figure 3: Unconditional samples from whole-image, fixed patch-stitching, fixed patch-averaging and PaDIS priors, shown in NI.

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